

Crash insurance and the G10 carry premium: theory notes

Version of 23 September 2026

These notes prove the results listed in `theory/README.md`. Throughout, Φ and φ are the standard normal distribution and density functions, $s = \sigma\sqrt{\tau}$, and for forward F and strike K

$$d_{\pm}(K) = \frac{\ln(F/K) \pm s^2/2}{s}.$$

We use the identity $K\varphi(d_-) = F\varphi(d_+)$ repeatedly.

R2. Arbitrage-free call prices

Proposition 1. *Let $c : [0, \infty) \rightarrow \mathbb{R}$ and $F > 0$. There is a random variable $X \geq 0$ with $\mathbb{E}X = F$ and $c(K) = \mathbb{E}(X - K)^+$ for all $K \geq 0$ if and only if (i) c is convex, (ii) $c(0) = F$, (iii) $c'_+(0) \geq -1$ and (iv) $c(K) \rightarrow 0$ as $K \rightarrow \infty$. The law of X is then unique, with distribution function $G = 1 + c'_+$ on $[0, \infty)$ and $\mathbb{P}(X = 0) = 1 + c'_+(0)$.*

Proof. Necessity. Convexity of c follows from convexity of $K \mapsto (x - K)^+$ and linearity of expectation, and $c(0) = \mathbb{E}X = F$. Since $0 \leq (X - K)^+ \leq X \in L^1$ and $(X - K)^+ \rightarrow 0$ pointwise, dominated convergence gives (iv). For $h > 0$, the difference quotient $[(X - K - h)^+ - (X - K)^+]/h$ is bounded by 1 in absolute value and converges to $-\mathbf{1}\{X > K\}$ as $h \downarrow 0$, so $c'_+(K) = -\mathbb{P}(X > K)$. In particular $c'_+(0) = -\mathbb{P}(X > 0) \geq -1$.

Sufficiency. A convex function on $[0, \infty)$ has a right derivative c'_+ that is nondecreasing and right-continuous, and c is absolutely continuous on compact intervals with $c(b) - c(a) = \int_a^b c'_+(u) du$. If $c'_+(K_0) > 0$ for some K_0 , then $c(K) \geq c(K_0) + c'_+(K_0)(K - K_0) \rightarrow \infty$, contradicting (iv); so $c'_+ \leq 0$, c is nonincreasing and, by (iv), $c \geq 0$. Let $L = \lim_{K \rightarrow \infty} c'_+(K) \leq 0$. Since c'_+ is nondecreasing, $c'_+ \leq L$ everywhere, so $c(K) \leq F + LK$; $L < 0$ would force $c < 0$ for large K . Hence $L = 0$.

Define $G(K) = 0$ for $K < 0$ and $G(K) = 1 + c'_+(K)$ for $K \geq 0$. Then G is nondecreasing and right-continuous, $G(0) = 1 + c'_+(0) \geq 0$ by (iii), and $G(K) \rightarrow 1$. It is therefore the distribution function of a random variable $X \geq 0$ (Lebesgue–Stieltjes construction). By Tonelli’s theorem,

$$\mathbb{E}(X - K)^+ = \int_K^\infty \mathbb{P}(X > u) du = \int_K^\infty (1 - G(u)) du = - \int_K^\infty c'_+(u) du = c(K) - \lim_{b \rightarrow \infty} c(b) = c(K),$$

and $\mathbb{E}X = c(0) = F$. Uniqueness holds because c determines c'_+ and hence G . $\square$

Remark 2. Extending c by $c(K) = F - K$ for $K < 0$, the second distributional derivative of c on $\mathbb{R}$ is the law of X ; the jump of c' at 0 carries the atom $\mathbb{P}(X = 0)$. On a finite strike grid the necessary conditions reduce to slopes in $[-1, 0]$ that are nondecreasing, which is the discrete test used in `src/qef/fx/arbitrage.py`.

R5(b). Premium-adjusted call delta

Let $D \in (0, 1]$ and $h(K) = D(K/F)\Phi(d_-(K))$ for $K > 0$. With spot delta, D is the base-currency discount factor; with forward delta, $D = 1$.

Lemma 3. *The inverse Mills ratio $\lambda(d) = \varphi(d)/\Phi(d)$ is strictly decreasing on $\mathbb{R}$, with $\lambda(d) \rightarrow \infty$ as $d \rightarrow -\infty$ and $\lambda(d) \rightarrow 0$ as $d \rightarrow \infty$.*

Proof. Differentiation gives $\lambda'(d) = -\lambda(d)[d + \lambda(d)]$, so it suffices that $d\Phi(d) + \varphi(d) > 0$. This function has derivative $\Phi(d) > 0$ and tends to 0 as $d \rightarrow -\infty$; hence $d\Phi(d) + \varphi(d) = \int_{-\infty}^d \Phi(u) du > 0$. The limits follow from $\Phi \rightarrow 0$, $\varphi \rightarrow 0$ and the Mills-ratio asymptotics $\Phi(d) \sim \varphi(d)/|d|$ as $d \rightarrow -\infty$. $\square$

Proposition 4. *The function h is continuous on $(0, \infty)$, tends to 0 as $K \downarrow 0$ and as $K \rightarrow \infty$, and satisfies $h < D$. There is a unique K^* such that h is strictly increasing on $(0, K^*]$ and strictly decreasing on $[K^*, \infty)$; $K^* = F \exp(-d^*s - s^2/2)$, where d^* is the unique solution of $\lambda(d) = s$. Each $\Delta \in (0, h(K^*))$ has exactly two preimages, one on each side of K^* .*

Proof. As $K \downarrow 0$, $d_- \rightarrow \infty$, so $h \rightarrow 0$. As $K \rightarrow \infty$, $d_- \rightarrow -\infty$ and, by the Mills-ratio asymptotics and the identity above, $K\Phi(d_-) \sim K\varphi(d_-)/|d_-| = F\varphi(d_+)/|d_-| \rightarrow 0$. Moreover $F\Phi(d_+) - K\Phi(d_-)$ is the undiscounted call price, which is positive, so $h < D\Phi(d_+) < D$.

Since $d'_-(K) = -1/(Ks)$,

$$h'(K) = \frac{D}{F} \left[\Phi(d_-) - \frac{\varphi(d_-)}{s} \right] = \frac{D\Phi(d_-)}{Fs} [s - \lambda(d_-)].$$

By the lemma, $s - \lambda(d)$ is strictly increasing in d and changes sign exactly once, at d^* . Because d_- is strictly decreasing in K , $h' > 0$ for $K < K^*$ and $h' < 0$ for $K > K^*$, where $d_-(K^*) = d^*$. The stated formula for K^* inverts d_- . The final claim follows from the intermediate value theorem and strict monotonicity on each side. $\square$

Remark 5. The market convention selects the preimage in $[K^*, \infty)$. At one month and 25Δ this root lies far to the right of K^* ; bracketing matters at long maturities and high volatilities.

R6. Diffusive null of the hedge cost

Let the undiscounted put value with lognormal forward G and volatility σ be

$$p(G) = K\Phi(-d_-(G)) - G\Phi(-d_+(G)), \quad d_{\pm}(G) = \frac{\ln(G/K) \pm s^2/2}{s}.$$

Using $K\varphi(d_-) = G\varphi(d_+)$, $p'(G) = -\Phi(-d_+(G))$.

Suppose that under $\mathbb{Q}$, S_τ is lognormal with mean F and log-variance s^2 , and that under $\mathbb{P}$ it is lognormal with the same log-variance and mean $Fe^{\lambda\tau}$ (no jumps, no volatility premium). A unit long forward costs nothing and has $\mathbb{P}$ -expected payoff $\mu_U = F(e^{\lambda\tau} - 1)$. Adding a put bought at its $\mathbb{Q}$ -value, with the premium financed to τ , gives $\mu_H = \mu_U + p(Fe^{\lambda\tau}) - p(F)$.

Proposition 6. *If $\lambda \neq 0$, the hedge-cost ratio $\theta_0 = 1 - \mu_H/\mu_U$ satisfies $\theta_0 = \Phi(-d_+(\xi))$ for some ξ between F and $Fe^{\lambda\tau}$, and*

$$|\theta_0 - \Phi(-d_1)| \leq \varphi(0) \frac{|\lambda|\sqrt{\tau}}{\sigma}, \quad d_1 = d_+(F).$$

Proof. By the mean value theorem, $p(Fe^{\lambda\tau}) - p(F) = -\Phi(-d_+(\xi)) F(e^{\lambda\tau} - 1)$ for some ξ between the two forwards, so $\theta_0 = -[p(Fe^{\lambda\tau}) - p(F)]/\mu_U = \Phi(-d_+(\xi))$. Since $|d_+(\xi) - d_+(F)| = |\ln(\xi/F)|/s \leq |\lambda|\tau/s$ and Φ is Lipschitz with constant $\varphi(0)$, the bound follows. $\square$

Remark 7. With no crash premium at all, hedging with a put of forward delta $-\Phi(-d_1)$ surrenders approximately that fraction of the carry premium: about 0.25 for a 25Δ put and 0.10 for a 10Δ put. This is the null against which the realised ratio θ_{UB} and the skew component of E3 are read.

R1. Domestic and foreign risk-neutral measures

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{Q}^d)$ satisfy the usual conditions, let B^d, B^f be strictly positive adapted money-market accounts with $B_0^d = B_0^f = 1$, and let $S > 0$ be adapted and càdlàg. Assume $Z_t = S_t B_t^f / (S_0 B_t^d)$ is a true $\mathbb{Q}^d$ -martingale and define $d\mathbb{Q}^f / d\mathbb{Q}^d = Z_T$ on $\mathcal{F}_T$.

Proposition 8. (a) $\mathbb{Q}^f$ is a probability measure equivalent to $\mathbb{Q}^d$. (b) For $\mathcal{F}_T$ -measurable $X \geq 0$ or $X \in L^1(\mathbb{Q}^f)$, $\mathbb{E}^{\mathbb{Q}^f}[X | \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t] / Z_t$. (c) An adapted process Y with $Y_t \in L^1(\mathbb{Q}^f)$ is a $\mathbb{Q}^f$ -martingale if and only if ZY is a $\mathbb{Q}^d$ -martingale; in particular a foreign price V^f satisfies: V^f / B^f is a $\mathbb{Q}^f$ -martingale if and only if $SV^f / B^d = S_0 Z (V^f / B^f)$ is a $\mathbb{Q}^d$ -martingale. (d) If $dS_t / S_t = (r_d - r_f) dt + \sigma dW_t$ under $\mathbb{Q}^d$ with constant coefficients, then $W_t^f = W_t - \sigma t$ is a $\mathbb{Q}^f$ -Brownian motion, $d(1/S_t) = (1/S_t)[(r_f - r_d) dt - \sigma dW_t^f]$, and $\mathbb{E}^{\mathbb{Q}^d} S_T = F$, $\mathbb{E}^{\mathbb{Q}^f}(1/S_T) = 1/F$ with $F = S_0 e^{(r_d - r_f)T}$.

Proof. (a) $Z_T > 0$ and $\mathbb{E}^{\mathbb{Q}^d} Z_T = Z_0 = 1$, so $\mathbb{Q}^f$ is a probability measure; positivity of Z_T gives equivalence. (b) For $\mathcal{F}_T$ -measurable $Y \geq 0$, the tower property and the martingale property give $\mathbb{E}^{\mathbb{Q}^f} Y = \mathbb{E}^{\mathbb{Q}^d}[Z_T Y] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y]$. For $A \in \mathcal{F}_t$,

$$\mathbb{E}^{\mathbb{Q}^f}[X \mathbf{1}_A] = \mathbb{E}^{\mathbb{Q}^d}[\mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t] \mathbf{1}_A] = \mathbb{E}^{\mathbb{Q}^f}\left[\frac{\mathbb{E}^{\mathbb{Q}^d}[Z_T X | \mathcal{F}_t]}{Z_t} \mathbf{1}_A\right],$$

which characterises the conditional expectation. (c) Apply (b) on $[s, t]$: $\mathbb{E}^{\mathbb{Q}^f}[Y_t | \mathcal{F}_s] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t | \mathcal{F}_s] / Z_s$, so Y is a $\mathbb{Q}^f$ -martingale exactly when $\mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t | \mathcal{F}_s] = Z_s Y_s$. (d) Here $Z_t = \exp(\sigma W_t - \sigma^2 t / 2)$, so Girsanov's theorem gives the first claim, and Itô's formula for $1/S$ gives the dynamics. The expectations follow from the lognormal law under each measure. $\square$

Remark 9. Jensen's inequality gives $\mathbb{E}^{\mathbb{Q}^d}(1/S_T) > 1/\mathbb{E}^{\mathbb{Q}^d} S_T = 1/F$. There is no contradiction with (d): the foreign investor's forward price is an expectation under $\mathbb{Q}^f$, not $\mathbb{Q}^d$ (Siegel's paradox). If Z is only a strict local martingale, $\mathbb{E} Z_T < 1$ and the construction fails (Carr, Fisher and Ruf, 2014).

R7. Entropy decomposition of currency premia

Fix t and write $\mathbb{E}_t$ for $\mathbb{E}[\cdot | \mathcal{F}_t]$. For a positive random variable M with $\mathbb{E}_t |\log M| < \infty$, the conditional entropy is $L_t(M) = \log \mathbb{E}_t M - \mathbb{E}_t \log M \geq 0$. Let $M, M^* > 0$ be $\mathcal{F}_{t+1}$ -measurable discount factors in L^2 that price domestic and foreign payoffs in their own currencies, with $\mathbb{E}_t M = e^{-r_t}$, $\mathbb{E}_t M^* = e^{-r_t^*}$, $\mathbb{E}_t |\log M|, \mathbb{E}_t |\log M^*| < \infty$ and $MS_{t+1}/S_t \in L^2$, where S is the domestic price of foreign currency. Write $m = \log M$, $\Delta s = \log(S_{t+1}/S_t)$, and $x_t = (f_t - s_t) - (r_t - r_t^*)$ for the covered-interest-parity basis.

Lemma 10. Suppose every foreign payoff $X^* \in L^2(\mathcal{F}_{t+1})$ is traded and priced consistently in both currencies, $S_t \mathbb{E}_t[M^* X^*] = \mathbb{E}_t[MS_{t+1} X^*]$. Then $M^* = MS_{t+1}/S_t$ almost surely.

Proof. The difference $D = M^* - MS_{t+1}/S_t$ lies in L^2 and satisfies $\mathbb{E}_t[DX^*] = 0$ for all $X^* \in L^2$. Taking $X^* = D$ gives $\mathbb{E}_t D^2 = 0$. $\square$

Proposition 11. *Under the lemma's conclusion, (i) $\mathbb{E}_t[\Delta s + r_t^* - r_t] = L_t(M) - L_t(M^*)$ and $\mathbb{E}_t[s_{t+1} - f_t] = L_t(M) - L_t(M^*) - x_t$. (ii) If $k(u) = \log \mathbb{E}_t e^{um}$ is finite near 0 and its Maclaurin series has radius of convergence greater than 1, then $L_t(M) = \sum_{j \geq 2} \kappa_{j,t}/j!$, where $\kappa_{j,t}$ are the conditional cumulants of m . (iii) For any gross return $R > 0$ with $\mathbb{E}_t[MR] = 1$ and $\mathbb{E}_t|\log R| < \infty$, $L_t(M) \geq \mathbb{E}_t \log R - r_t$.*

Proof. (i) By definition, $\mathbb{E}_t m = -r_t - L_t(M)$ and $\mathbb{E}_t m^* = -r_t^* - L_t(M^*)$. The lemma gives $\Delta s = m^* - m$, so $\mathbb{E}_t \Delta s = r_t - r_t^* + L_t(M) - L_t(M^*)$. The second identity follows from $s_{t+1} - f_t = \Delta s - (f_t - s_t) = \Delta s - (r_t - r_t^*) - x_t$. (ii) $L_t(M) = k(1) - k'(0)$ with $k'(0) = \kappa_{1,t}$, and $k(1) = \sum_{j \geq 1} \kappa_{j,t}/j!$ under the stated convergence. (iii) By Jensen's inequality, $0 = \log \mathbb{E}_t[MR] \geq \mathbb{E}_t m + \mathbb{E}_t \log R$, so $L_t(M) = -r_t - \mathbb{E}_t m \geq \mathbb{E}_t \log R - r_t$. $\square$

Remark 12. The identity in (i) needs only that some pair of discount factors prices bonds and exchange rates consistently; completeness makes the pair unique. Complete markets imply cross-country discount-factor correlations that are hard to reconcile with consumption data (Brandt, Cochrane and Santa-Clara, 2006). Options on S reveal the risk-neutral law of $\Delta s = m^* - m$, not the separate cumulants of m and m^* , so they constrain but do not identify the split of $L_t(M) - L_t(M^*)$ across cumulant orders. Part (iii) is the entropy bound of Bansal and Lehmann (1997); hedged and unhedged carry returns bound the same $L_t(M)$.

Results still to be written

R3, R4, R5(c), R8 and the Garman–Kohlhagen PDE, as stated in `theory/README.md`.